

Code-Layout, Readability and Reusability

Date	30 june 2026
Team ID	XXXXXX
Project Name	Opticrop-Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout	yes	ollowed PEP 8 guidelines for Python and used Prettier for HTML/JS.
2	Proper File Structure	Files and folders are logically organize	Yes	Organized into templates/, static/, models/, and app.py logic.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	sed descriptive names like soil_ph, nitrogen_lvl, and crop_yield.
4	Function / Method Names	Functions are descriptively named	Yes	dded Python docstrings and inline comments for complex ML transformations.
5	Code Comments	Inline and block comments explain log	yes	Added Python docstrings and inline comments for complex ML transformations.
6	Modular Design	Code is split into reusable functions/m	yes	Separated ML inference, database interactions, and web routing.
7	No Redundant Code	Duplicate or unused code is removed	yes	Refactored repeated API calls into a single reusable service class.
8	Error Handling	Exceptions and errors are handled gra	yes	Implemented try-except blocks for external API failures and invalid form inputs.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	ML engine	Python / Scikit-learn (Reused in Scenarios 1 & 2)	-	high

2	Weather api	Python / Requests (Reused when climate data is missing)	-	High
3	datavalidatiot	Python (Reused across all form submissions	-	High
4	Ui layouts	Navbar/Footer)HTML / CSS / Jinja2 (Included in all web views)	-	high
5	Db_connector	-	-	high

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean MVC-like separation of frontend, routing, and ML models. Readability
Readability	4	Clear naming conventions; some complex pandas aggregations require slightly more documentation. Reusability
Reusability	5	Core Machine Learning logic is highly modular and decoupled from the web framework. Documentation / Comments
Documentation / Comments	3	Docstrings are present; API documentation could be expanded further. Overall Score
Overall Score	4.5	Excellent overall code health, ready for scalable deployment.